



Today And Beyond

The current decade is seeing technology driving us rather than the other way round—and we're happy slaves

It's not hard to sum up this, the current decade, in two words: "Smaller, faster!" The '00s is the decade of miniaturisation and speed. The century, of course, belongs to nanotechnology.

Already by 2000, we'd lost our innocence. No more the tinkering around with toys and making inventions of them. We've been going aggressively ahead, looking at the basic laws of physics and asking, "Why not?"

"Miniaturisation" is not restricted to gadgets: we're also getting to the nanoscale. A great example is the DVD to blue-laser transition: we packed more onto a DVD than we could on a CD, so find new materials! Invent better microscopes! Dig out more of nature's secrets! Do whatever, and figure out *just how densely those bits can be packed!* It was in this spirit of scientific aggressiveness that HD-DVD and Blu-ray were born—as with almost everything else that's happening in tech today: processors, RAM, and all the rest.

Early this decade was born the Pentium 4. AMD came into its own this decade, and became a true competitor after long years of lagging behind. The gigahertz races had begun. (An aside: AMD reached the 1 GHz finish line first with its 1 GHz Athlon, just days before Intel released the 1 GHz Pentium III.) Then someone got the not-so-original idea that if one processor could do this much

work, why not employ two or more—and thus came in dual-core and multi-core: the future of desktop computing.

Meanwhile, the graphics processor was becoming less specialised in its capabilities and more flexible in its nature. Thus was born the idea of the Graphics Processing Unit, or the GPU. The first GPU was the NVIDIA GeForce 256, launched in August of 1999.

If the GPU primarily enhanced the gaming experience, what was needed was better sound—and starting off from the simple SoundBlaster sound cards of yore, we're moving on to immersive sound experiences. The x.x speaker thing is for the most part a phenomenon of this decade: 4.1, 5.1, now 7.1 and even 9.1. (Maybe there's a 11.1 set—only we don't know!) Similarly with displays: with so many visual bytes ready to be delivered, we needed better things to display them. So CRTs became better, LCD technology went mainstream, and OLED and plasma displays came in. Sure, LCDs are still weak when it comes to games, but do you really believe they won't be on par by, say, 2012?

Games have thus far been driving much of the PC consumer industry. We now have physics processors for games. The PS3 uses the Cell processor, dubbed a super-computer in its own right.

While the Internet in the '90s was all about expansion, it's now

about broadband: people are clamouring for more. More speed, more content, more services. They want *all* their data to come in through one pipe, too. That's why IPTV and VoIP are taking off: one line, many services—WWW, TV, phone, movies. And all high-definition, high-fidelity.

(Pause for breath.) Think about how much broadband has proliferated: in fact, in the developed world, everyone has it, so it almost makes no sense to call it by a special name. From 2 to 5 to 10 Mbps—to 50 and even 100 Mbps in some countries. (Off the top of our heads; there might even be 1 Gbps in some places!)

Not only do we want more, we want it along, wherever we go. Yes, the mobile revolution: it all happened so fast, the timeline is a series of months rather than years. Sometime between 1995 and 2000, it happened in India: by 2003 or so, everybody had a cell phone. In the developed world, kids as young as 8 and 10 years old have cell phones. (Younger kids are tending to get RFID-tagged in countries such as Japan.) And once you've got a gadget, why not pack more into it? Naturally, we don't need to tell you about when cell phones became phones cum PDAs cum MP3 players and radio tuners and what not.

It doesn't seem to make sense to call it a "phone" anymore; it's just "the Gadget." These Gadgets have processing power, and they're

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